The $RO(C_q)$-Graded Cohomology of the Complete Flag Variety of $\mathbb{C}[C_q]$

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Abstract

Let q be an odd prime and let $G = C_q$ act on the variety X of complete flags in its complex regular representation. The fixed flags of G are those of the maximal torus containing it, and every invariant curve of this torus is G-invariant. In $RO(G)$-graded Bredon cohomology with coefficients in $\mathbb{F}_q$, however, the Euler classes of all nontrivial weights are zero divisors and generate the same ideal, so the hypotheses of the known theorems of Goresky–Kottwitz–MacPherson type fail. Let R be the cohomology of a point, λ the real representation underlying a faithful character, and a its Euler class. We show that the cohomology of X is a free R-module on Schubert classes σ_w of degree $\ell(w)\lambda$. In this basis restriction to the fixed flags has matrix $M\,\mathrm{diag}(a^{\ell(w)})$, where M is invertible over $\mathbb{F}_q$. This determines the kernel, which is nonzero and of square zero, the image, and the quotient of the solutions of the curve congruences by the image. We also give a presentation of the ring, the Mackey functor structure, and the mod q cohomology of the orbit space X/G.

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1 Introduction

Let an algebraic torus $\mathbb{T}$ act on a smooth complex projective variety with finitely many fixed points and finitely many invariant curves. Such a variety is equivariantly formal, and Goresky, Kottwitz and MacPherson showed that restriction to the fixed points embeds its equivariant cohomology, in characteristic zero, as the ring of tuples of polynomials that agree along each invariant curve modulo the weight of that curve [11, Theorems 1.2.2 and 14.1]. Harada, Henriques and Holm [12] extended this to other equivariant cohomology theories, under the hypothesis that the Euler classes of the weights at each fixed point are non-zero-divisors and pairwise relatively prime. Flag varieties are the basic examples. Here we replace the torus acting on a flag variety by a finite cyclic subgroup that has the same fixed points and still preserves the invariant curves of the torus. Both hypotheses on the Euler classes then fail, and we determine what replaces the GKM description.

Let q be an odd prime, let $G = C_q = \langle g \rangle$, and let $\chi\colon G \to \mathbb{C}^\times$ be the character with $\chi(g) = \exp(2\pi i/q)$. The regular representation $V = \mathbb{C}[G]$ is the sum of the characters $\chi^0, \dots, \chi^{q-1}$, each occurring once. A subspace of V is therefore G-invariant exactly when it is a sum of eigenlines, and on the variety $X = \mathrm{Fl}(V)$ of complete flags the cyclic group G has the same fixed points as the diagonal torus $\mathbb{T} \subset GL(V)$, namely the $q!$ coordinate flags. We call them the fixed flags. The $\mathbb{T}$-invariant curves joining them, the root curves, are G-invariant as well. We study X in $RO(G)$-graded Bredon cohomology with coefficients in the constant Mackey functor $\underline{\mathbb{F}_q}$, written $H^\star_G(-)$, and we let $R = H^\star_G(\mathrm{pt})$ denote the coefficient ring.

In $H^*(BG; \mathbb{F}_q) = \mathbb{F}_q[t] \otimes \Lambda(s)$ the first Chern class of χ^b is bt, so every weight of $\mathbb{T}$ that occurs on X restricts to a unit multiple of the single class $t = c_1(\chi)$. The congruence along a root curve then asks for one factor of t, whereas a Schubert class of length d restricts to each fixed flag as a multiple of t^d. The curves cannot detect this higher divisibility. In Bredon cohomology, which we also call the genuine theory to distinguish it from Borel cohomology, there is a second effect. The role of t is played by the Euler class $a \in R^\lambda$ of the real representation λ underlying χ, and a is a zero divisor in R. This gives restriction to the fixed flags a kernel.

Results

We fix eigenlines $E_j \cong \chi^{j-1}$ in V for $1 \le j \le q$ and write p_v, for $v \in S_q$, for the coordinate flag whose ith subspace is $E_{v_1} \oplus \cdots \oplus E_{v_i}$. Permutations are written in one-line notation, $\ell(w)$ is the number of inversions of w, and τ_{kl} is the transposition of k and l. On X we have the tautological flag $F_1 \subset \cdots \subset F_q = V$ and the line bundles $L_i = F_i/F_{i-1}$. Besides a, the ring R contains an orientation class $u \in R^{\lambda-2}$, which becomes invertible in Borel cohomology. We write $X_{hG} = EG \times_G X$ for the Borel construction, $z_i = c_1(L_i^\vee) \in H^2(X_{hG}; \mathbb{F}_q)$, and ϕ_X for the comparison map from $H^\star_G(X)$ to $RO(G)$-graded Borel cohomology; these are recalled in Sections 2 and 4.

Theorem 1.1. (a) *There are unique classes $x_1, \dots, x_q \in H_G^\lambda(X)$ with $\phi_X(x_i) = uz_i$, and they induce an isomorphism of R-algebras*

$$H_G^\star(X) \cong \frac{R[x_1, \dots, x_q]}{(e_1(x), \dots, e_{q-2}(x),\ e_{q-1}(x) + a^{q-1},\ e_q(x))}, \tag{1.1}$$

where e_k is the kth elementary symmetric polynomial. Equivalently, $\prod_{i=1}^q (T - x_i) = T^q - a^{q-1}T$.

(b) *As an R-module, $H_G^\star(X)$ is free on the monomials $x_1^{b_1} \cdots x_{q-1}^{b_{q-1}}$ with $0 \le b_i \le q - i$. It is also free on the Schubert classes $\sigma_w \in H_G^{\ell(w)\lambda}(X)$, $w \in S_q$, defined in Section 5.3.*

Relative to the classical presentation of the cohomology of a flag variety, the only relation that changes is $e_{q-1}(x) = 0$, which becomes $e_{q-1}(x) = -a^{q-1}$. The reason is that the Chern roots of the characters of V are the classes ct with $c \in \mathbb{F}_q$, and $\prod_{c \in \mathbb{F}_q}(T - ct) = T^q - t^{q-1}T$. Products with the Schubert classes of degree λ are given by a Monk formula in which a enters as well (Proposition 5.1).

Let $i\colon X^G \hookrightarrow X$ be the inclusion of the fixed flags, so that

$$i^*\colon H_G^\star(X) \longrightarrow H_G^\star(X^G) = \prod_{v \in S_q} R.$$

The group $R^{d\lambda}$ is spanned by a^d for $d \ge 0$, so there are scalars $M_{v,w} \in \mathbb{F}_q$ with $\sigma_w|_{p_v} = M_{v,w}\, a^{\ell(w)}$. They are values of double Schubert polynomials; see (6.1). For each pair $k < l$ the fixed flags p_v and $p_{v\tau_{kl}}$ are joined by a root curve, a projective line on which G acts through a nontrivial character. Restricting to these curves shows that every $f = (f_v)$ in the image of i^* satisfies $f_v - f_{v\tau_{kl}} \in aR$. Let $\Gamma \subset \prod_v R$ be the R-module of all tuples satisfying these congruences, and for $d \ge 0$ let $J_d = \operatorname{Ann}_R(a^d)$.

Theorem 1.2. *The matrix $M = (M_{v,w})$ is invertible over $\mathbb{F}_q$, and in the Schubert basis i^* is given by $M \operatorname{diag}(a^{\ell(w)})$. It follows that:*

(a) *The kernel of i^* is $\bigoplus_{w \ne \mathrm{id}} J_{\ell(w)}\sigma_w$. It is nonzero, and the product of any two of its elements vanishes.*

(b) *A tuple $f \in \prod_v R$ lies in the image of i^* if and only if $(M^{-1}f)_w \in a^{\ell(w)}R$ for every $w \in S_q$. Hence i^* becomes an isomorphism after inverting a.*

(c) *The image of i^* is contained in Γ, and*

$$\Gamma / \operatorname{im} i^* \cong \bigoplus_{\ell(w) \ge 2} aR/a^{\ell(w)}R \tag{1.2}$$

as graded R-modules.

We call the quotient (1.2) the defect of the root curves. An explicit formula for M^{-1} is given in Proposition 6.1; for instance, the condition in (b) for the longest permutation says that $\sum_v \operatorname{sgn}(v) f_v$ is divisible by $a^{q(q-1)/2}$. The simplest tuple in Γ that is not a restriction has a single nonzero entry, equal to a, at the longest permutation (Remark 6.3). The defect is thus already nonzero in degree λ, although i^* is injective in every degree $d\lambda$, where the coefficients of the Schubert classes are scalar multiples of powers of a. On the other hand, the coefficients of the kernel lie in the negative cone $N \subset R$, the ideal of elements annihilated by a power of a (Section 2). This ideal has square zero and satisfies $a^d N = N$ for every d, so every tuple with entries in N is a restriction and N does not

contribute to the defect. The defect is in fact the same as in Borel cohomology with t in place of a, whereas the kernel vanishes there (Section 6.3).

We also determine the Mackey functor structure. The Schubert classes form a basis at both orbits G/G and G/e, and the transfer is computed from the transfer on the coefficients (Proposition 7.1). Taking integer degrees yields the cohomology of the orbit space. For $n \geq 2$ the dimension of $H^n(X/G; \mathbb{F}_q)$ is the number of permutations of length at least $\lceil n/2 \rceil$, and all products of classes of positive degree vanish (Corollary 7.2). Section 8 writes everything out for $q = 3$.

Method

The argument has three steps. First, the Bruhat cells of X are complex representations of G in which every character is nontrivial. With coefficients in $\underline{\mathbb{F}_q}$ all the spheres of such characters become equivalent after smashing with $H = H\underline{\mathbb{F}_q}$, and the vanishing of $R^{1-d\lambda}$ for $d \geq 0$ then forces the cellular filtration to split: $H \wedge X_+$ is a wedge of the suspensions $\Sigma^{\ell(w)\lambda} H$ (Proposition 3.1). Second, for any space with such a splitting the comparison with Borel cohomology is an isomorphism in the degrees $d\lambda$, and the whole $RO(G)$-graded ring is recovered from the even Borel cohomology by regrading and extending scalars along $\mathbb{F}_q[t] \to R$, $t \mapsto a$ (Proposition 4.2). Third, the Borel construction of X is a flag bundle over BG, so its cohomology and its Schubert calculus are classical.

The classes x_i are defined through the comparison with Borel cohomology and not as Euler classes. At p_v the fiber of L_i is the character $\chi^{v_i - 1}$, which is trivial for $v_i = 1$ and nontrivial otherwise. As a real representation the fiber is therefore trivial at some fixed flags and irreducible at others, so a construction using a Thom class of L_i in a fixed degree is not immediate. The same variation occurs in the tower of projective bundles from X down to $\mathbb{P}(V)$: over the point E_j of $\mathbb{P}(V)$ the tautological quotient bundle has fiber V/E_j, which contains the trivial character unless $j = 1$. The projective space computations of [4] concern projectivizations of representations over a point, so we use the projective bundle formula only after passing to Borel cohomology.

Related work

The computations closest to ours concern projective spaces. Lewis [18] treated complex projective spaces with linear $\mathbb{Z}/p$-actions and Burnside ring coefficients. More recently Basu, Dey and Karmakar showed that, for a cyclic group and integral coefficients, the projective space of a multiple of the regular representation splits after smashing with the Eilenberg–Mac Lane spectrum into a wedge of representation suspensions [4, Theorem 4.7]. Proposition 3.1 is the analogue for complete flags and coefficients $\underline{\mathbb{F}_q}$, where the comparison of the spheres of different characters [4, Propositions 2.10–2.11] makes all summands suspensions by multiples of one representation. The coefficient ring R is taken from Basu and Ghosh [2, §3.2], who also prove a freeness theorem for a class of complexes built from representation cells [2, Theorem 4.12].

For $G = C_2$ and coefficients $\underline{\mathbb{F}_2}$ there is a substantial literature on Grassmannians: Kronholm's freeness theorem [17], for which Hogle and May gave a corrected proof and an extension to complexes of finite type [15], Dugger's computation of the rings of the infinite real Grassmannians [9], and Hogle's determination of the additive structure for families of finite real and complex Grassmannians [14]. For $q = 2$, which is excluded here, the flag variety $\mathrm{Fl}(\mathbb{C}[C_2])$ is the projective line $\mathbb{P}(\mathbb{C}[C_2])$, which is covered by [14, Theorem 8.1]; it is a single root curve, and the analogue of Theorem 1.2 reduces to the two-point computation at the beginning of Section 6.3.

On the side of GKM theory, the theorem of Harada, Henriques and Holm [12, Theorem 3.1] is the natural reference point, and Section 6.3 compares it with Theorem 1.2 in detail. We also use their description of the Schubert cell filtration of a flag variety [12, §5]. With coefficients in

$\mathbb{F}_p$, Franz and Puppe prove that the Atiyah–Bredon sequence of a torus action is exact when the equivariant cohomology is free over $H^*(B\mathbb{T};\mathbb{F}_p)$ and the filtration by orbit dimension agrees with the filtration by orbit size for the maximal p-torus [10, Theorem 2.1]. Their result is about torus actions. For the cyclic group itself every orbit outside X^G has size q, so the filtration by orbit size has no intermediate stage, and the root curves have to be put in by hand. Theorem 1.2 says that they do not suffice.

Organization

Section 2 recalls the coefficient ring. Section 3 proves the splitting of $H \wedge X_+$ and Section 4 the comparison with Borel cohomology. Theorem 1.1 is proved in Section 5 and Theorem 1.2 in Section 6. Section 7 treats the Mackey functor structure and the orbit space, and Section 8 the case $q = 3$. In Section 9 we discuss what changes for the cyclic group of order q^2, where we have only partial information. The appendix contains the computation of the transfer on the coefficients.

2 The coefficient ring

2.1 Conventions

We work in the stable homotopy category of genuine G-spectra indexed on a complete universe [19]. For a G-space Y and $\alpha \in RO(G)$ we put

$$H^\alpha_G(Y) = [\Sigma^\infty Y_+, \Sigma^\alpha H]^G = \pi^G_{-\alpha} F(Y_+, H), \qquad H = H\underline{\mathbb{F}}_q.$$

These groups are the values at the orbit G/G of Mackey functors, which we denote by underlining: $\underline{H}^\alpha_G(Y)$ has the value $H^\alpha_G(Y)$ at G/G and the nonequivariant cohomology of Y at G/e, and its structure maps are $\mathrm{res} = \mathrm{res}^G_e$ and $\mathrm{tr} = \mathrm{tr}^G_e$. For a point we obtain the coefficient ring, with $R^\alpha = H^\alpha_G(\mathrm{pt})$, and the coefficient Mackey functor $\underline{R}^\alpha$. We refer to res as restriction to the trivial subgroup, to distinguish it from restriction to the fixed flags, which is the pullback i^*. Modules over H and maps between them are understood in the homotopy category of H-modules, and $[-,-]_H$ denotes morphisms there.

Put $m = (q-1)/2$. The nontrivial irreducible real representations of G are the realifications $\lambda_r = (\chi^r)_{\mathbb{R}}$ for $1 \le r \le m$, and $RO(G)$ is free on $1, \lambda_1, \dots, \lambda_m$. We write $\lambda = \lambda_1$. For $m < b < q$ complex conjugation identifies the realification of χ^b with λ_{q-b}. Since q is odd, every λ_r is of complex type, and the sign in the commutation rule for products in $H^\star_G$ depends only on the coefficients of the trivial representation in the two degrees. Classes whose degree is a multiple of λ are therefore central.

For a representation W with $W^G = 0$ the Euler class $a_W \in R^W$ is the class of the inclusion $S^0 \to S^W$, and we write $a = a_\lambda$.

2.2 Comparing the spheres S^{λ_r}

After smashing with H the spheres S^{λ_r} all become equivalent. With integer coefficients this is [3, Proposition 3.6]; see also [4, Propositions 2.10–2.11]. With coefficients in a field the proof is short, and we include it because the explicit map fixes our normalizations.

Lemma 2.1. *For $1 \le r \le m$ let $[r]\colon S^\lambda \to S^{\lambda_r}$ be the one-point compactification of $z \mapsto z^r$. Then $1 \wedge [r]\colon H \wedge S^\lambda \to H \wedge S^{\lambda_r}$ is an equivalence of H-modules.*

Proof. The map $[r]$ is equivariant, has degree r on underlying spheres, and is the identity on the fixed points $S^0 = \{0, \infty\}$. Let C be the cofiber of $1 \wedge [r]$. On underlying spectra $1 \wedge [r]$ is multiplication by r on $\Sigma^2 H\mathbb{F}_q$, which is an equivalence because r is prime to q. Hence the underlying spectrum of C is contractible, $EG_+ \wedge C$ is contractible, and the isotropy separation sequence gives $C \simeq \widetilde{EG} \wedge C$. The fixed points of $\widetilde{EG} \wedge C$ are the geometric fixed points $\Phi^G C$, the cofiber of $\Phi^G(1 \wedge [r])$. This map is the identity of $\Phi^G H$, since $[r]$ is the identity on fixed spheres. So the homotopy groups of C vanish at both subgroups of G, and C is contractible. □

Under the identification $[\Sigma^\lambda H, \Sigma^{\lambda_r} H]_H = R^{\lambda_r - \lambda}$ we let ε_r be the class of $r^{-1}(1 \wedge [r])$. It is a unit, and $\varepsilon_1 = 1$. Composing $[r]$ with $S^0 \to S^\lambda$ gives $S^0 \to S^{\lambda_r}$, so

$$a_{\lambda_r} = r\, \varepsilon_r\, a \qquad (1 \le r \le m). \tag{2.1}$$

Two consequences will be used repeatedly. First, for every nontrivial character χ^b, with realification ρ,

$$H \wedge S^\rho \simeq \Sigma^\lambda H \qquad \text{and} \qquad a_\rho R = aR. \tag{2.2}$$

Second, every element of $RO(G)$ has a unique expression $n + d\lambda + \sum_{r \ge 2} c_r(\lambda_r - \lambda)$, and multiplication by $\prod_r \varepsilon_r^{c_r}$ identifies $R^{n+d\lambda}$ with the group in that degree. Writing $R_{1,\lambda} = \bigoplus_{n,d \in \mathbb{Z}} R^{n+d\lambda}$ and $\mathcal{L} = \mathbb{F}_q[\varepsilon_2^{\pm 1}, \dots, \varepsilon_m^{\pm 1}]$, we get

$$R = \mathcal{L} \otimes_{\mathbb{F}_q} R_{1,\lambda}. \tag{2.3}$$

The units ε_r thus only move classes between degrees with the same underlying dimension and the same fixed-point dimension, and computations in R reduce to computations in $R_{1,\lambda}$ followed by extension of scalars to $\mathcal{L}$. For $q = 3$ there are no such units. For $q = 5$ there is one, ε_2, with $a_{\lambda_2} = 2\varepsilon_2 a$, and a class of degree λ such as x_i in Theorem 1.1 gives the class $\varepsilon_2 x_i$ of degree λ_2. A different choice of the units ε_r would only change (2.1) by scalars.

2.3 The positive and the negative cone

The ring $R_{1,\lambda}$ was computed by Basu and Ghosh [2, Proposition 3.4]. It is a direct sum

$$R_{1,\lambda} = P \oplus N_{1,\lambda}, \qquad P = \mathbb{F}_q[a, u] \otimes \Lambda(\kappa), \qquad |a| = \lambda, \quad |u| = \lambda - 2, \quad |\kappa| = \lambda - 1, \tag{2.4}$$

of a subring P, the positive cone, and an ideal $N_{1,\lambda}$, the negative cone. The latter arises from the Tate square as the desuspension of the cokernel of $P[u^{-1}] \oplus P[a^{-1}] \to P[u^{-1}, a^{-1}]$. It therefore has an $\mathbb{F}_q$-basis of classes $\nu_{j,k}$ and $\mu_{j,k}$, for $j, k \ge 1$, corresponding to $u^{-j}a^{-k}$ and $\kappa u^{-j} a^{-k}$, in degrees

$$|\nu_{j,k}| = 1 + 2j - (j+k)\lambda, \qquad |\mu_{j,k}| = 2j + (1 - j - k)\lambda, \tag{2.5}$$

and P acts on it by

$$\begin{gathered} a\nu_{j,k} = \nu_{j,k-1}, \quad u\nu_{j,k} = \nu_{j-1,k}, \quad a\mu_{j,k} = \mu_{j,k-1}, \quad u\mu_{j,k} = \mu_{j-1,k}, \\ \kappa\nu_{j,k} = \mu_{j,k}, \qquad \kappa\mu_{j,k} = 0, \end{gathered} \tag{2.6}$$

where a symbol with an index equal to zero is read as zero. We put $N = \mathcal{L} \otimes_{\mathbb{F}_q} N_{1,\lambda}$, so that $R = (\mathcal{L} \otimes P) \oplus N$.

Lemma 2.2. (a) *$R^{d\lambda} = \mathbb{F}_q a^d$ for $d \ge 0$ and $R^{d\lambda} = 0$ for $d < 0$. Moreover $R^{1-d\lambda} = 0$ for all $d \ge 0$.*

(b) *Multiplication by a is injective on $\mathcal{L}\otimes P$ and maps N onto N. For $d\geq 1$,*

$$J_d = \mathrm{Ann}_R(a^d) = \mathcal{L}\otimes_{\mathbb{F}_q} \mathrm{span}_{\mathbb{F}_q}\{\nu_{j,k},\mu_{j,k} : j\geq 1,\ 1\leq k\leq d\}.$$

(c) $N^2 = 0$.

Proof. Degrees of the form $d\lambda$ or $1-d\lambda$ contain no λ_r with $r\geq 2$, so by (2.3) only $R_{1,\lambda}$ contributes to (a). A monomial $a^i u^j \kappa^\eta$ of P has degree $(i+j+\eta)\lambda - (2j+\eta)$, and by (2.5) the coefficient of the trivial representation in the degree of a basis element of $N_{1,\lambda}$ is at least 2. That coefficient is thus never 1, and it is 0 only for the powers of a. This gives (a). For (b), $\mathcal{L}\otimes P$ is a polynomial ring over $\mathcal{L}\otimes\Lambda(\kappa)$ in which a is a variable, while on N multiplication by a lowers the second index. The annihilator of a^d is therefore contained in N and is spanned as stated. For (c), let $n, n'\in N$. By (b) some power a^d annihilates n, and $n' = a^d n''$ for some $n''\in N$. Since a is central, $nn' = (a^d n)n'' = 0$. □

2.4 Borel cohomology of a point

Let $R_h = \pi^G_{-\star}F(EG_+, H)$ be the $RO(G)$-graded Borel cohomology of a point and $\phi\colon R\to R_h$ the map induced by $EG_+\to S^0$. In integer degrees R_h is $H^*(BG;\mathbb{F}_q) = \mathbb{F}_q[t]\otimes\Lambda(s)$, where $t = c_1(\chi)$ and $s\in H^1(BG;\mathbb{F}_q)$ is chosen with Bockstein t. The orientation class u maps to the Thom class of λ, a unit of R_h, for which we keep the same letter, and we do the same for the units ε_r. According to [2, §3.2],

$$R_h = \mathbb{F}_q[t]\otimes\Lambda(s)\otimes\mathbb{F}_q[u^{\pm1},\varepsilon_2^{\pm1},\dots,\varepsilon_m^{\pm1}],\qquad \phi(a)=ut,\quad \phi(\kappa)=us,\quad \phi(N)=0. \tag{2.7}$$

In other words $R_h = R[u^{-1}]$, and the negative cone dies because it is u-torsion.

Lemma 2.3. *For every $d\in\mathbb{Z}$ the map $\phi\colon R^{d\lambda}\to R_h^{d\lambda}$ is an isomorphism.*

Proof. A monomial $u^k t^h s^\eta \prod_r \varepsilon_r^{c_r}$ of R_h has degree $(2h+\eta-2k)+k\lambda+\sum_r c_r(\lambda_r-\lambda)$. This equals $d\lambda$ only if all c_r vanish, $k=d$, $\eta=0$ and $h=d$. So $R_h^{d\lambda}$ is spanned by $(ut)^d = \phi(a^d)$ if $d\geq 0$ and is zero otherwise, and the claim follows from Lemma 2.2(a). □

2.5 Restriction and transfer on the coefficients

At the orbit G/e the coefficient ring is the cohomology of a point in the nonequivariant sense, graded over $RO(G)$. It is concentrated in degrees of underlying dimension zero, where it is

$$\underline{R}^\star(G/e) = \mathbb{F}_q[\bar u^{\pm1},\bar\varepsilon_2^{\pm1},\dots,\bar\varepsilon_m^{\pm1}],\qquad \bar u = \mathrm{res}(u),\quad \bar\varepsilon_r = \mathrm{res}(\varepsilon_r). \tag{2.8}$$

The classes a and κ map to zero under res for dimensional reasons. For $\mathbf{c} = (c_2,\dots,c_m)\in\mathbb{Z}^{m-1}$ we write $\varepsilon^{\mathbf{c}} = \prod_r \varepsilon_r^{c_r}$ and $\bar\varepsilon^{\mathbf{c}} = \mathrm{res}(\varepsilon^{\mathbf{c}})$.

Lemma 2.4. *The map* res *vanishes on N. After multiplying all the classes $\nu_{j,k}$ and $\mu_{j,k}$ by a common nonzero scalar, which does not affect* (2.6)*, the transfer is given by*

$$\mathrm{tr}(\bar u^{\,n}\bar\varepsilon^{\mathbf{c}}) = \begin{cases} 0, & n\geq 0,\\ \varepsilon^{\mathbf{c}}\mu_{-n,1}, & n<0.\end{cases} \tag{2.9}$$

The proof is a computation with lens spaces and is given in Appendix A. From now on the generators of N are normalized as in the lemma.

3 Schubert cells and the splitting

Choose a generator $\mathbf{e}_j$ of each eigenline E_j, so that $g\mathbf{e}_j = \chi(g)^{j-1}\mathbf{e}_j$. We compose permutations as functions, $(vw)(i) = v(w(i))$, so that right multiplication by τ_{kl} exchanges the entries in positions k and l. We write $s_i = \tau_{i,i+1}$ for the adjacent transpositions, $w \le v$ for the Bruhat order, and w_0 for the longest permutation. As noted in the introduction,

$$X^G = \{p_w : w \in S_q\}, \qquad (p_w)_i = E_{w_1} \oplus \cdots \oplus E_{w_i}. \tag{3.1}$$

Let $\mathcal{B} \subset GL(V)$ be the group of matrices that are upper triangular in the basis $\mathbf{e}_1, \dots, \mathbf{e}_q$. Its orbits on X are the Schubert cells $\mathcal{B}p_w$. A flag in $\mathcal{B}p_w$ has a unique basis in echelon form

$$\mathbf{e}_{w_i} + \sum_{\substack{j>i \\ w_j<w_i}} c_{ji}\,\mathbf{e}_{w_j} \qquad (1 \le i \le q),$$

whose first i vectors span its ith subspace, and the coefficients c_{ji} are coordinates on the cell. Applying g and rescaling the basis vectors multiplies c_{ji} by $\chi(g)^{w_j-w_i}$. As a G-space the cell is therefore the complex representation

$$W_w = \bigoplus_{\substack{i<j \\ w_i>w_j}} \chi^{w_j-w_i}, \tag{3.2}$$

of complex dimension $\ell(w)$. The exponents lie strictly between $-q$ and 0, so no character in W_w is trivial. The splitting below rests on this property.

The closure of $\mathcal{B}p_w$ is the union of the cells $\mathcal{B}p_v$ with $v \le w$ [8, Proposition 1.2.1(iii)]. The subspaces

$$X_{\le d} = \bigcup_{\ell(w)\le d} \mathcal{B}p_w, \qquad 0 \le d \le \tbinom{q}{2},$$

are therefore closed and G-invariant, and $X_{\le d}$ is obtained from $X_{\le d-1}$ by attaching the unit discs of the representations W_w with $\ell(w) = d$ along $\mathbb{T}$-equivariant maps of their boundary spheres [12, §5]. Restricting the action to G, we obtain cofiber sequences of based G-spaces

$$(X_{\le d-1})_+ \longrightarrow (X_{\le d})_+ \longrightarrow \bigvee_{\ell(w)=d} S^{W_w} \qquad (d \ge 1). \tag{3.3}$$

Proposition 3.1. *There is an equivalence of H-modules*

$$H \wedge X_+ \simeq \bigvee_{w\in S_q} \Sigma^{\ell(w)\lambda} H. \tag{3.4}$$

Proof. Put $U_d = H \wedge (X_{\le d})_+$. By (2.2) and (3.2), $H \wedge S^{W_w} \simeq \Sigma^{\ell(w)\lambda}H$, so smashing (3.3) with H gives a cofiber sequence of H-modules

$$U_{d-1} \longrightarrow U_d \longrightarrow Q_d \xrightarrow{\delta_d} \Sigma U_{d-1}, \qquad Q_d = \bigvee_{\ell(w)=d} \Sigma^{d\lambda} H.$$

Assume inductively that U_{d-1} is a wedge of modules $\Sigma^{j\lambda}H$ with $0 \le j < d$; this holds for $U_0 = H$. The components of δ_d then lie in

$$[\Sigma^{d\lambda}H, \Sigma^{j\lambda+1}H]_H = R^{1-(d-j)\lambda},$$

which is zero by Lemma 2.2(a). Hence $\delta_d = 0$ and $U_d \simeq U_{d-1} \vee Q_d$. □

The splitting is thus forced by degrees alone, and no attaching map has to be computed. The coefficients $\underline{\mathbb{F}_q}$ are used both to identify $H \wedge S^{W_w}$ with $\Sigma^{\ell(w)\lambda}H$ and to obtain the vanishing of $R^{1-d\lambda}$.

4 Comparison with Borel cohomology

For a G-space Y let $H_h^\star(Y) = \pi_{-\star}^G F(EG_+ \wedge Y_+, H)$ be its $RO(G)$-graded Borel cohomology, and let

$$\phi_Y \colon H_G^\star(Y) \longrightarrow H_h^\star(Y)$$

be the map induced by $EG_+ \to S^0$. It is a ring homomorphism, natural in Y, and for $Y = \mathrm{pt}$ it is the map ϕ of Section 2. In integer degrees $H_h^n(Y) = H^n(Y_{hG}; \mathbb{F}_q)$, where $Y_{hG} = EG \times_G Y$, and since u is a unit of R_h we have $H_h^{d\lambda}(Y) = u^d H^{2d}(Y_{hG}; \mathbb{F}_q)$ for all d. From now on cohomology of a Borel construction is taken with coefficients in $\mathbb{F}_q$.

Proposition 4.1. *Let Y be a G-space such that, for some finite set I and integers $d_k \geq 0$,*

$$H \wedge Y_+ \simeq \bigvee_{k \in I} \Sigma^{d_k \lambda} H \qquad \text{as } H\text{-modules.} \tag{4.1}$$

Then for every $d \in \mathbb{Z}$ the comparison map is an isomorphism

$$\phi_Y \colon H_G^{d\lambda}(Y) \xrightarrow{\cong} H_h^{d\lambda}(Y) = u^d H^{2d}(Y_{hG}). \tag{4.2}$$

Proof. By adjunction and (4.1),

$$F(Y_+, H) \simeq F_H(H \wedge Y_+, H) \simeq \bigvee_{k \in I} \Sigma^{-d_k \lambda} H$$

as H-modules, where F_H denotes the function spectrum of H-modules. The map $E \to F(EG_+, E)$ is natural in the G-spectrum E. Applied to this decomposition, it identifies ϕ_Y in degree $d\lambda$ with the direct sum of the maps $\phi \colon R^{(d-d_k)\lambda} \to R_h^{(d-d_k)\lambda}$, which are isomorphisms by Lemma 2.3. □

Let $\mathcal{H}_\lambda(Y) = \bigoplus_{d \geq 0} H_G^{d\lambda}(Y)$. It is a commutative graded algebra over $\mathcal{H}_\lambda(\mathrm{pt}) = \mathbb{F}_q[a]$. Under (4.1) the groups $H_G^{d\lambda}(Y)$ with $d < 0$ vanish, so $\mathcal{H}_\lambda(Y)$ contains every class whose degree is a multiple of λ. Let $H^{\mathrm{ev}}(Y_{hG}) = \bigoplus_{d \geq 0} H^{2d}(Y_{hG})$. Proposition 4.1 says that

$$\theta_Y \colon H^{\mathrm{ev}}(Y_{hG}) \longrightarrow \mathcal{H}_\lambda(Y), \qquad \theta_Y(b) = \phi_Y^{-1}(u^d b) \quad \text{for } b \in H^{2d}(Y_{hG}), \tag{4.3}$$

is an isomorphism of rings. It takes H^{2d} to degree $d\lambda$, sends t to a by (2.7), and is natural for G-maps between spaces satisfying (4.1).

Proposition 4.2. *Let Y satisfy* (4.1)*. Then multiplication induces an isomorphism of R-algebras $R \otimes_{\mathbb{F}_q[a]} \mathcal{H}_\lambda(Y) \to H_G^\star(Y)$. Consequently*

$$R \otimes_{\mathbb{F}_q[t]} H^{\mathrm{ev}}(Y_{hG}) \xrightarrow{\cong} H_G^\star(Y), \qquad c \otimes b \longmapsto c\,\theta_Y(b), \tag{4.4}$$

where R is an $\mathbb{F}_q[t]$-algebra via $t \mapsto a$. This isomorphism is natural for G-maps between spaces satisfying (4.1)*.*

Proof. The decomposition of $F(Y_+, H)$ in the proof of Proposition 4.1 provides classes $\beta_k \in H_G^{d_k\lambda}(Y)$ that form a basis of $H_G^\star(Y)$ as an R-module. In degree $d\lambda$ this reads $H_G^{d\lambda}(Y) = \bigoplus_k R^{(d-d_k)\lambda}\beta_k$, so by Lemma 2.2(a) the same classes form a basis of $\mathcal{H}_\lambda(Y)$ over $\mathbb{F}_q[a]$. The multiplication map therefore carries a basis to a basis. It is multiplicative because $\mathcal{H}_\lambda(Y)$ is central in $H_G^\star(Y)$. The second statement follows by composing with θ_Y. □

Both X, by Proposition 3.1, and the finite set X^G satisfy (4.1), so these results apply to X, to X^G and to the inclusion between them. The next section applies Proposition 4.2 to the even Borel cohomology of X.

5 The ring structure

5.1 Borel cohomology of X

Let $\mathcal{V} = EG \times_G V$ be the vector bundle over BG associated with V. The Borel construction X_{hG} is the bundle of complete flags in $\mathcal{V}$, and the tautological line bundles on it are the Borel constructions of the L_i. Recall that $z_i = c_1(L_i^\vee) \in H^2(X_{hG})$. The Chern roots of $\mathcal{V}^\vee$ are the classes $c_1(E_j^\vee) = r_j t$ with

$$r_j = -(j-1) \in \mathbb{F}_q \qquad (1 \le j \le q). \tag{5.1}$$

As j varies, r_j runs through all of $\mathbb{F}_q$, so

$$\prod_{j=1}^{q} (T - r_j t) = T^q - t^{q-1} T. \tag{5.2}$$

The cohomology of a flag bundle is obtained by iterating the projective bundle formula [1, §4.4]: it is free over the cohomology of the base on the monomials $z_1^{b_1} \cdots z_{q-1}^{b_{q-1}}$ with $0 \le b_i \le q-i$, and the only relations are those expressing that $\prod_i (T - z_i)$ is the Chern polynomial of $\mathcal{V}^\vee$. Since the projective bundle formula holds with arbitrary coefficients, it does not matter that q divides the order of the Weyl group. With (5.2) we get

$$H^*(X_{hG}) = \frac{(\mathbb{F}_q[t] \otimes \Lambda(s))[z_1, \dots, z_q]}{(e_1(z), \dots, e_{q-2}(z),\ e_{q-1}(z) + t^{q-1},\ e_q(z))}, \tag{5.3}$$

and since s is the only generator of odd degree, $H^{\mathrm{ev}}(X_{hG})$ is the quotient of $\mathbb{F}_q[t, z_1, \dots, z_q]$ by the same relations.

5.2 Generators and relations

Proof of Theorem 1.1(a) *and of the first half of* (b). Put $x_i = \theta_X(z_i)$. This is the unique class in $H_G^\lambda(X)$ with $\phi_X(x_i) = u z_i$, because ϕ_X is injective in degree λ. By Proposition 4.2 and the description of $H^{\mathrm{ev}}(X_{hG})$ just given, $H_G^\star(X)$ is obtained by extending scalars along $t \mapsto a$ from the quotient of $\mathbb{F}_q[t, z_1, \dots, z_q]$ by the relations in (5.3), with z_i corresponding to x_i. This is the presentation (1.1), and the monomial basis of $H^{\mathrm{ev}}(X_{hG})$ over $\mathbb{F}_q[t]$ becomes the monomial basis over R. □

Applying the naturality of θ to the inclusion of a fixed flag, and noting that L_i restricts to E_{v_i} at p_v, we find

$$z_i|_{p_v} = r_{v_i} t, \qquad x_i|_{p_v} = r_{v_i} a = -(v_i - 1)\, a. \tag{5.4}$$

5.3 Schubert classes

Let $Z = (Z_1, \dots, Z_q)$ and $B = (B_1, \dots, B_q)$ be two sets of variables. The double Schubert polynomials $\mathfrak{S}_w(Z; B) \in \mathbb{Z}[Z, B]$, $w \in S_q$, are defined by

$$\mathfrak{S}_{w_0}(Z; B) = \prod_{i+j \le q} (Z_i - B_j), \qquad \mathfrak{S}_{ws_i} = \partial_i \mathfrak{S}_w \quad \text{if } \ell(ws_i) < \ell(w), \tag{5.5}$$

where $\partial_i f = (f - s_i f)/(Z_i - Z_{i+1})$ is the divided difference operator of Bernstein, Gel'fand and Gel'fand [5] and s_i exchanges Z_i and Z_{i+1}. We use the following standard facts [1, §§10.6, 10.7 and 10.10].

(S1) $\mathfrak{S}_w$ is homogeneous of degree $\ell(w)$. Moreover $\mathfrak{S}_{\mathrm{id}} = 1$ and $\mathfrak{S}_{s_i} = \sum_{j\le i}(Z_j - B_j)$.

(S2) Let $\mathcal{A} = \mathbb{Z}[Z,B]/(e_k(Z) - e_k(B) : 1 \le k \le q)$. The classes of the $\mathfrak{S}_w$, $w \in S_q$, form a basis of $\mathcal{A}$ as a module over $\mathbb{Z}[B]$.

(S3) In $\mathcal{A}$,
$$\mathfrak{S}_{s_i}\mathfrak{S}_w = \sum_{j\le i}(B_{w_j} - B_j)\,\mathfrak{S}_w + \sum_{\substack{k\le i<l\\ \ell(w\tau_{kl})=\ell(w)+1}} \mathfrak{S}_{w\tau_{kl}}.$$

(S4) For $v \in S_q$ the specialization $\mathfrak{S}_w(B_{v_1},\dots,B_{v_q};B)$ vanishes unless $w \le v$, and
$$\mathfrak{S}_w(B_{w_1},\dots,B_{w_q};B) = \prod_{\substack{i<j\\ w_i>w_j}} (B_{w_i} - B_{w_j}).$$

Geometrically, $\mathcal{A}$ is the $\mathbb{T}$-equivariant cohomology of $\mathrm{Fl}(\mathbb{C}^q)$ with integer coefficients, where $Z_i = c_1^{\mathbb{T}}(L_i^\vee)$ and $B_j = c_1^{\mathbb{T}}(E_j^\vee)$, and $\mathfrak{S}_w$ is the class of the closure Ω_w of the orbit of p_w under the lower triangular matrices [1, §10.6]. Statement (S2) is the fact that these classes form a basis over $H^*(B\mathbb{T};\mathbb{Z})$, (S3) is the equivariant Monk rule, and (S4) records that $p_v \in \Omega_w$ only for $v \ge w$, together with the Euler class of the normal space to Ω_w at p_w.

By Theorem 1.1(a) and (5.2) we have $\prod_i(T - x_i) = \prod_j(T - r_j a)$ in $H^\star_G(X)[T]$. There is therefore a ring homomorphism
$$\Psi\colon \mathcal{A} \longrightarrow H^\star_G(X), \qquad Z_i \longmapsto x_i, \quad B_j \longmapsto r_j a, \tag{5.6}$$
and we define
$$\sigma_w = \Psi(\mathfrak{S}_w) = \mathfrak{S}_w(x_1,\dots,x_q;r_1a,\dots,r_qa) \in H_G^{\ell(w)\lambda}(X). \tag{5.7}$$

Proof of the rest of Theorem 1.1. The presentation (1.1) says that Ψ induces an isomorphism $R \otimes_{\mathbb{Z}[B]} \mathcal{A} \cong H^\star_G(X)$, where $\mathbb{Z}[B] \to R$ sends B_j to $r_j a$. By (S2) the classes σ_w form a basis over R, which completes (b). □

By (S1) the Schubert classes of degree λ are
$$\sigma_{s_i} = x_1 + \cdots + x_i + \tbinom{i}{2}\, a \qquad (1 \le i < q), \tag{5.8}$$
and their products with arbitrary Schubert classes are given by the following Monk formula.

Proposition 5.1. *For $1 \le i < q$ and $w \in S_q$,*
$$\sigma_{s_i}\sigma_w = \Big(\sum_{j=1}^{i}(j - w_j)\Big)a\,\sigma_w + \sum_{\substack{k\le i<l\\ \ell(w\tau_{kl})=\ell(w)+1}} \sigma_{w\tau_{kl}}, \tag{5.9}$$
where the integer coefficient is reduced modulo q.

Proof. Apply Ψ to (S3). The coefficient $B_{w_j} - B_j$ is sent to $(r_{w_j} - r_j)a = (j - w_j)a$. □

More generally, the coefficient of σ_z in a product $\sigma_v\sigma_w$ lies in $R^{(\ell(v)+\ell(w)-\ell(z))\lambda}$, so by Lemma 2.2(a) every structure constant of the Schubert basis is a scalar multiple of a power of a. The negative cone enlarges the ring but creates no new structure constants.

Remark 5.2. Under the comparison map, $\phi_X(\sigma_w) = u^{\ell(w)}S_w$ with $S_w = \mathfrak{S}_w(z;r_1t,\dots,r_qt)$, and S_w is the class of Ω_w in the Borel cohomology of X. This is the sense in which we call σ_w a Schubert class. The arguments below use only the definition (5.7) and do not require a construction of σ_w as a fundamental class of Ω_w in genuine cohomology.

6 Restriction to the fixed flags

6.1 The restriction matrix

For $v \in S_q$ write $r_v = (r_{v_1}, \dots, r_{v_q})$ and $r = r_{\mathrm{id}}$, with r_j as in (5.1). Define

$$M_{v,w} = \mathfrak{S}_w(r_v; r) \in \mathbb{F}_q \qquad (v, w \in S_q). \tag{6.1}$$

We also put $\ell_0 = \ell(w_0) = \binom{q}{2}$ and $\Delta_v = \prod_{i<j}(v_j - v_i) \in \mathbb{F}_q^\times$, so that $\Delta_v = \mathrm{sgn}(v)\Delta_{\mathrm{id}}$.

Proposition 6.1. (a) $\sigma_w|_{p_v} = M_{v,w}\, a^{\ell(w)}$ *for all* $v, w \in S_q$.

(b) $M_{v,w} = 0$ *unless* $w \le v$, *and* $M_{w,w} = \prod_{i<j,\, w_i>w_j}(w_j - w_i) \neq 0$. *Hence* M *is invertible.*

(c) $M_{v,\mathrm{id}} = 1$ *and* $M_{v,s_i} = \sum_{j\le i}(j - v_j)$ *for all* v.

(d) *The inverse of* M *is given by*

$$(M^{-1})_{w,v} = (-1)^{\ell_0 - \ell(w)}\, \frac{M_{w_0 v,\, w_0 w}}{\Delta_v}. \tag{6.2}$$

In particular $(M^{-1})_{w_0,v} = \mathrm{sgn}(v)/\Delta_{\mathrm{id}}$.

Proof. By (5.4), restriction to p_v composed with Ψ sends Z_i to $r_{v_i}a$ and B_j to $r_j a$. Part (a) follows since $\mathfrak{S}_w$ is homogeneous of degree $\ell(w)$, and (c) follows from (S1). Part (b) is (S4) evaluated at $B = r$. The factors $w_j - w_i$ are nonzero in $\mathbb{F}_q$ because $0 < |w_j - w_i| < q$, and a matrix that is triangular with respect to a linear extension of the Bruhat order and has nonzero diagonal entries is invertible.

For (d), consider the polynomials in Z

$$P_w(Z) = \mathfrak{S}_w(Z; r), \qquad Q_w(Z) = \mathfrak{S}_{w_0 w}(Z; r_{w_0}),$$

of degrees $\ell(w)$ and $\ell_0 - \ell(w)$. Thus $P_w(r_v) = M_{v,w}$. To evaluate Q_w we use that double Schubert polynomials are unchanged when a constant is added to all the variables Z_i and B_j, as one sees from (5.5). Since $r_{w_0(j)} = (1-q) - r_j$ and hence $r_{v_i} = (1-q) - r_{(w_0 v)_i}$, translation by $q-1$ and homogeneity give

$$Q_w(r_v) = \mathfrak{S}_{w_0 w}(-r_{w_0 v}; -r) = (-1)^{\ell_0 - \ell(w)} M_{w_0 v,\, w_0 w}. \tag{6.3}$$

The right-hand side of (6.2) is therefore $Q_w(r_v)/\Delta_v$, and we have to show that

$$c_{w,b} = \sum_{v \in S_q} \frac{Q_w(r_v)\, P_b(r_v)}{\Delta_v}$$

equals 1 if $b = w$ and 0 otherwise. Suppose first that $\ell(b) < \ell(w)$. Then $Q_w P_b$ has degree less than ℓ_0. Its antisymmetrization $\sum_v \mathrm{sgn}(v)(Q_w P_b)(Z_{v_1}, \dots, Z_{v_q})$ is divisible by the Vandermonde determinant, which has degree ℓ_0, so it vanishes. Evaluating at $Z = r$ and using $\Delta_v = \mathrm{sgn}(v)\Delta_{\mathrm{id}}$ gives $c_{w,b} = 0$. Suppose now that $\ell(b) \ge \ell(w)$. By (b), a term of the sum is nonzero only if $b \le v$ and $w_0 w \le w_0 v$, that is, only if $b \le v \le w$. This forces $b = v = w$, so $c_{w,b} = 0$ for $b \neq w$, and $c_{w,w} = Q_w(r_w)P_w(r_w)/\Delta_w$. By (b) the second factor is the product of $w_j - w_i$ over the inversions of w. The inversions of $w_0 w$ are the non-inversions of w, so by (6.3) and (b) the first factor is the product of $w_j - w_i$ over the pairs $i < j$ with $w_i < w_j$. Together they give Δ_w, and $c_{w,w} = 1$. The last assertion of (d) is the case $w = w_0$ of (6.2), since $M_{w_0 v,\mathrm{id}} = 1$. □

Formula (6.2) expresses the duality between the Schubert polynomials of w and $w_0 w$ under the operator ∂_{w_0}, which is the antisymmetrization used in the proof divided by the Vandermonde determinant. All entries of M can be written as sums over reduced subwords by the formula of Billey [6, Theorem 4]; we shall only need the properties listed in the proposition.

Remark 6.2. One can work with the monomial basis of Theorem 1.1(b) instead. Writing $|\mathbf{b}| = \sum_i b_i$, the restriction of $x^{\mathbf{b}}$ to p_v is $\mathsf{E}_{v,\mathbf{b}}\, a^{|\mathbf{b}|}$ with $\mathsf{E}_{v,\mathbf{b}} = \prod_i r_{v_i}^{b_i}$, and E is again invertible. Indeed, by Lemma 2.2(a) the two bases are related by $x^{\mathbf{b}} = \sum_w c_{\mathbf{b},w}\, a^{|\mathbf{b}|-\ell(w)}\sigma_w$ with scalars $c_{\mathbf{b},w} \in \mathbb{F}_q$. Both are homogeneous bases of $\mathcal{H}_\lambda(X)$ over $\mathbb{F}_q[a]$, so the determinant of this change of basis is a nonzero constant and the matrix $(c_{\mathbf{b},w})$ is invertible. Restricting to the fixed flags gives $\mathsf{E}_{v,\mathbf{b}} = \sum_w M_{v,w} c_{\mathbf{b},w}$. We prefer the Schubert basis because M is triangular and has the explicit inverse (6.2).

6.2 Kernel and image

Proof of Theorem 1.2(a) *and* (b). Invertibility of M and the matrix of i^* are contained in Proposition 6.1. For $h = \sum_w c_w \sigma_w$ with $c_w \in R$ this means

$$(M^{-1} i^* h)_w = a^{\ell(w)} c_w \qquad (w \in S_q). \tag{6.4}$$

So h lies in the kernel exactly when $c_w \in J_{\ell(w)}$ for all w, where $J_0 = 0$, and f lies in the image exactly when each $(M^{-1}f)_w$ is divisible by $a^{\ell(w)}$. After inverting a every J_d becomes zero and every $a^{\ell(w)}$ a unit, so i^* becomes an isomorphism.

By Lemma 2.2(b) the coefficients of an element of the kernel lie in N. The Schubert classes are central and $N^2 = 0$, so the product of two elements of the kernel is zero. Finally $a\mu_{1,1} = 0$ by (2.6), so $\mu_{1,1}\sigma_{s_1}$ is a nonzero element of the kernel. Its degree is $2 - \lambda + \lambda = 2$. □

The cokernel of i^* can be read off from (6.4) in the same way: it is isomorphic to $\bigoplus_{w \neq \mathrm{id}} R/a^{\ell(w)}R$. In a degree $d\lambda$ the coefficients c_w lie in $R^{(d-\ell(w))\lambda}$, on which multiplication by $a^{\ell(w)}$ is injective by Lemma 2.2(a), so i^* is injective in these degrees. At the other extreme, $a^d N = N$ for every d by Lemma 2.2(b). If f has entries in N, then so does $M^{-1}f$, and each $(M^{-1}f)_w$ is divisible by $a^{\ell(w)}$. Every tuple with entries in N is therefore in the image of i^*.

6.3 Root curves and the defect

Let ρ be the realification of a nontrivial character ψ of G. The sphere S^ρ has the two fixed points 0 and ∞, and

$$\operatorname{im}\bigl(H^\star_G(S^\rho) \to H^\star_G(\{0,\infty\}) = R \times R\bigr) = \{(f_0, f_\infty) : f_0 - f_\infty \in aR\}. \tag{6.5}$$

Indeed, $H^\star_G(S^\rho)$ is free over R on the unit and on the generator of the reduced group $\widetilde{H}^\rho_G(S^\rho)$, where S^ρ is based at ∞. The latter restricts to 0 at ∞ and, by the definition of the Euler class, to a_ρ at 0. So the image consists of the pairs with $f_0 - f_\infty \in a_\rho R$, and $a_\rho R = aR$ by (2.2).

Now let $v \in S_q$ and $k < l$. Replacing the summand E_{v_k} of the subspaces $(p_v)_i$, $k \le i < l$, by an arbitrary line in $E_{v_k} \oplus E_{v_l}$ defines a G-invariant curve in X through p_v and $p_{v\tau_{kl}}$. It is isomorphic to $\mathbb{P}(E_{v_k} \oplus E_{v_l})$, hence to S^ρ for $\psi = \chi^{v_l - v_k}$, with the two fixed flags as poles. These are the root curves. By (6.5) the tuples that extend over all of them form the module Γ of the introduction,

$$\Gamma = \bigl\{ f \in \textstyle\prod_v R : f_v - f_{v\tau_{kl}} \in aR \text{ for all } v \in S_q \text{ and } k < l \bigr\}.$$

Proof of Theorem 1.2(c). Since transpositions generate S_q, a tuple belongs to Γ if and only if all its entries are congruent modulo aR. Writing $\mathbf{1}$ for the tuple with all entries 1, this means

$$\Gamma = R\,\mathbf{1} + a\textstyle\prod_v R. \tag{6.6}$$

The entries of M are scalars, so M^{-1} preserves $a\prod_v R$, and $M^{-1}\mathbf{1}$ is the basis vector indexed by id because the corresponding column of M is $\mathbf{1}$ by Proposition 6.1(c). Therefore

$$M^{-1}\Gamma = R \oplus \bigoplus_{w\neq \mathrm{id}} aR, \qquad M^{-1}\operatorname{im} i^* = \bigoplus_w a^{\ell(w)}R,$$

the second equality being Theorem 1.2(b). The first module contains the second, and the quotient is (1.2). □

The group G has many more invariant curves than $\mathbb{T}$. For example, by (3.2) the cell of w_0 contains the character χ^{-1} with multiplicity $q-1$, and every line through the origin in the corresponding coordinate subspace is G-invariant. The closure of such a line is the image of a sphere S^ρ whose poles map to fixed flags, so by (6.5) it imposes a congruence modulo aR between two entries of a tuple. By (6.6) this congruence already holds on Γ, and enlarging the set of curves would not change Theorem 1.2(c).

The same computation can be made in Borel cohomology. The ring $H^{\mathrm{ev}}(X_{hG})$ restricts injectively to $\prod_v \mathbb{F}_q[t]$ with matrix $M\operatorname{diag}(t^{\ell(w)})$, the root curves impose congruences modulo t, and the quotient of the solutions of these congruences by the image is $\bigoplus_{\ell(w)\geq 2} t\mathbb{F}_q[t]/t^{\ell(w)}\mathbb{F}_q[t]$. The defect (1.2) is the base change of this quotient. Indeed, N is contained in a^dR for every d, so

$$aR/a^dR \cong a\bar{R}/a^d\bar{R} \cong \bar{R}\otimes_{\mathbb{F}_q[t]} \left(t\mathbb{F}_q[t]/t^d\mathbb{F}_q[t]\right), \qquad \bar{R} = R/N = \mathcal{L}\otimes P,$$

where $\bar{R}$ is free over $\mathbb{F}_q[t]$ via $t\mapsto a$. Every summand of (1.2) is nonzero, since $a\notin a^2R$ by Lemma 2.2(a).

Remark 6.3. Let $f\in\prod_v R$ have entry a at w_0 and entry 0 elsewhere. Then $f\in\Gamma$ by (6.6). By Proposition 6.1(d), $(M^{-1}f)_{w_0} = \operatorname{sgn}(w_0)\,a/\Delta_{\mathrm{id}}$. This is a nonzero element of R^λ, whereas the part of $a^{\ell_0}R$ in degree λ is $a^{\ell_0}R^{(1-\ell_0)\lambda} = 0$ by Lemma 2.2(a). Hence f satisfies all congruences along root curves but is not the restriction of a class on X.

We can now make the comparison with the theorem of Harada, Henriques and Holm precise. In their setting the cell of p_w contributes one Euler class for each character in W_w, and they assume that these classes are non-zero-divisors and pairwise relatively prime, which means that an element divisible by each of them is divisible by their product. This is the last of their four assumptions [12, Assumption 4], and under these assumptions their theorem says that restriction to the fixed points is injective and that its image is the analogue of Γ [12, Theorem 3.1]. In our case each of these Euler classes generates the ideal aR by (2.2). The first half of the assumption fails because $a\mu_{1,1} = 0$, and this accounts for the kernel. The second half fails at every cell of length $d\geq 2$, where a is divisible by each Euler class but not by their product, which generates a^dR; this accounts for the summand aR/a^dR of the defect.

7 The Mackey functor and the orbit space

7.1 Restriction and transfer

The value of $\underline{H}^\star_G(X)$ at G/e is the nonequivariant cohomology of X, graded over $RO(G)$. Let $y_i = c_1(L_i^\vee)\in H^2(X;\mathbb{F}_q)$. By (2.8) and the classical presentation of the cohomology of a flag variety

[7, Proposition 31.1],

$$\underline{H}^\star_G(X)(G/e) = \frac{\mathbb{F}_q[\bar{u}^{\pm 1}, \bar{\varepsilon}_2^{\pm 1}, \dots, \bar{\varepsilon}_m^{\pm 1}][y_1, \dots, y_q]}{(e_1(y), \dots, e_q(y))}, \tag{7.1}$$

with y_i in degree 2. The comparison map commutes with restriction to the trivial subgroup, for which Borel cohomology is ordinary cohomology, and z_i restricts to y_i on the fiber X of $X_{hG} \to BG$. Hence $\mathrm{res}(x_i) = \bar{u}\, y_i$, and since $\mathrm{res}(a) = 0$,

$$\bar{\sigma}_w := \mathrm{res}(\sigma_w) = \bar{u}^{\,\ell(w)}\, \mathfrak{S}_w(y; 0). \tag{7.2}$$

The polynomials $\mathfrak{S}_w(y; 0)$ are the ordinary Schubert polynomials. By (S2), specialized at $B = 0$, they form a basis of $H^*(X; \mathbb{F}_q)$.

Proposition 7.1. *For every $\alpha \in RO(G)$,*

$$\underline{H}^\alpha_G(X)(G/G) = \bigoplus_{w \in S_q} R^{\alpha - \ell(w)\lambda}\, \sigma_w, \qquad \underline{H}^\alpha_G(X)(G/e) = \bigoplus_{w \in S_q} \underline{R}^{\alpha - \ell(w)\lambda}(G/e)\, \bar{\sigma}_w, \tag{7.3}$$

and in terms of these decompositions

$$\mathrm{res}(c\, \sigma_w) = \mathrm{res}(c)\, \bar{\sigma}_w, \qquad \mathrm{tr}(b\, \bar{\sigma}_w) = \mathrm{tr}(b)\, \sigma_w \tag{7.4}$$

for $c \in R$ and $b \in \underline{R}^\star(G/e)$. Moreover $\mathrm{res} \circ \mathrm{tr} = 0$ *and* $\mathrm{tr} \circ \mathrm{res} = 0$.

Proof. The first decomposition is Theorem 1.1(b), and the second follows from (7.1) and (7.2) because $\bar{u}$ is a unit. The map res is a ring homomorphism, which gives the first formula in (7.4); the second is Frobenius reciprocity, $\mathrm{tr}(b\, \mathrm{res}(\sigma_w)) = \mathrm{tr}(b)\, \sigma_w$. For the last statement, $\mathrm{tr} \circ \mathrm{res}$ is multiplication by $\mathrm{tr}(1) = q = 0$. The composite $\mathrm{res} \circ \mathrm{tr}$ is the sum of the maps induced by the elements of G on $\underline{H}^\star_G(X)(G/e)$. Each of them is the identity, because G acts on X and on the representation spheres through connected groups, and so $\mathrm{res} \circ \mathrm{tr} = q = 0$ as well. □

Combined with Lemma 2.4, the proposition determines $\underline{H}^\star_G(X)$ completely. For example, $\mathrm{tr}(\bar{u}^{-j}\bar{\sigma}_w) = \mu_{j,1}\sigma_w$ for $j \geq 1$, while $\mathrm{tr}(\bar{u}^{\,n}\bar{\sigma}_w) = 0$ for $n \geq 0$; for $q = 5$ one has in addition classes such as $\mathrm{tr}(\bar{u}^{-1}\bar{\varepsilon}_2\bar{\sigma}_w) = \varepsilon_2\mu_{1,1}\sigma_w$ in the degrees involving λ_2.

The kernel of restriction to the fixed flags behaves differently at the two orbits. At G/e this map sends every y_i to zero, so its kernel is the ideal $(y_1, \dots, y_q)$, in which products such as y_1^2 are nonzero. The vanishing of products in Theorem 1.2(a) is special to the orbit G/G.

7.2 The orbit space

For a finite G-CW complex, Bredon cohomology in integer degrees with coefficients in a constant Mackey functor is the cohomology of the orbit space. The flag variety X is a compact smooth G-manifold and hence a finite G-CW complex [16], so the integer-graded part of Theorem 1.1 computes $H^*(X/G; \mathbb{F}_q)$.

Corollary 7.2. *We have $H^0(X/G; \mathbb{F}_q) = \mathbb{F}_q$ and $H^1(X/G; \mathbb{F}_q) = 0$, and for $n \geq 2$*

$$\dim_{\mathbb{F}_q} H^n(X/G; \mathbb{F}_q) = \#\{w \in S_q : \ell(w) \geq \lceil n/2 \rceil\}. \tag{7.5}$$

The product of any two classes of positive degree is zero.

Proof. By Theorem 1.1(b), $H^n_G(X) = \bigoplus_w R^{n-\ell(w)\lambda}\sigma_w$. For $n = 0$ only the summand of $\sigma_{\mathrm{id}} = 1$ is nonzero, by Lemma 2.2(a). Let $n \geq 1$ and $\ell \geq 0$. The degree $n - \ell\lambda$ lies in $\mathbb{Z} \oplus \mathbb{Z}\lambda$, so by (2.3) the group $R^{n-\ell\lambda}$ is the part of $R_{1,\lambda} = P \oplus N_{1,\lambda}$ in that degree. The positive cone does not contribute, because its monomials $a^i u^j \kappa^\eta$ have degree $(i + j + \eta)\lambda - (2j + \eta)$, whose integer part is not positive. By (2.5), $R^{n-\ell\lambda}$ is therefore spanned by $\mu_{j,\ell-j+1}$ if $n = 2j$ and $\ell \geq j$, by $\nu_{j,\ell-j}$ if $n = 2j + 1$ and $\ell \geq j + 1$, and is zero in all other cases. This gives the dimensions. All classes of positive degree have coefficients in N, and $N^2 = 0$. □

For $q = 3$ the Betti numbers of X/G in degrees $0, \ldots, 6$ are $1, 0, 5, 3, 3, 1, 1$.

8 The case $q = 3$

For $q = 3$ we have $RO(G) = \mathbb{Z}\{1, \lambda\}$, there are no units ε_r, and $(r_1, r_2, r_3) = (0, -1, 1)$. Write $x = x_1$ and $y = x_2$. Eliminating $x_3 = -x - y$ from (1.1), we obtain

$$H^\star_G(X) = \frac{R[x, y]}{(x^3 - a^2 x,\ x^2 + xy + y^2 - a^2)}, \tag{8.1}$$

with basis $1, x, y, x^2, xy, x^2 y$ over R. The first relation says that x is a root of $T^3 - a^2 T$, and the second is $e_2(x) + a^2 = 0$ up to sign.

We index Schubert classes by one-line notation. From $\mathfrak{S}_{321} = (Z_1 - B_1)(Z_1 - B_2)(Z_2 - B_1)$ and the divided differences one finds

$$\sigma_{213} = x, \quad \sigma_{132} = x + y + a, \quad \sigma_{231} = xy, \quad \sigma_{312} = x(x + a), \quad \sigma_{321} = xy(x + a), \tag{8.2}$$

and the Monk formula (5.9) gives, for instance,

$$\sigma_{213}^2 = \sigma_{312} - a\,\sigma_{213}, \qquad \sigma_{132}^2 = \sigma_{231} - a\,\sigma_{132}, \qquad \sigma_{213}\sigma_{132} = \sigma_{231} + \sigma_{312}.$$

The first and third of these are immediate from (8.2), and the second is equivalent to the quadratic relation in (8.1).

With the permutations ordered as $123, 213, 132, 231, 312, 321$, both for the fixed flags and for the Schubert classes, substituting $x_i|_{p_v} = r_{v_i} a$ into (8.2) gives

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}, \qquad M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ -1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix} \tag{8.3}$$

over $\mathbb{F}_3$, in agreement with (6.2). Theorem 1.2 now reads as follows. The kernel of i^* is

$$J_1\sigma_{213} \oplus J_1\sigma_{132} \oplus J_2\sigma_{231} \oplus J_2\sigma_{312} \oplus J_3\sigma_{321},$$

and a tuple $f = (f_1, \ldots, f_6)$, indexed in the order above, is in the image of i^* if and only if

$$\begin{aligned} f_1 - f_2 &\in aR, \\ f_1 - f_3 &\in aR, \\ f_1 + f_2 - f_3 - f_4 &\in a^2 R, \\ f_1 - f_2 + f_3 - f_5 &\in a^2 R, \\ -f_1 + f_2 + f_3 - f_4 - f_5 + f_6 &\in a^3 R. \end{aligned}$$

The root curves only give that all f_i are congruent modulo aR, and

$$\Gamma/\operatorname{im} i^* \cong (aR/a^2R)^{\oplus 2} \oplus aR/a^3R.$$

The last of the five conditions is the alternating sum of Proposition 6.1(d), and the tuple $(0,0,0,0,0,a)$ of Remark 6.3 violates it. In positive integer degrees i^* is zero, because $R^n = 0$ for $n > 0$, so all of $H^n(X/G;\mathbb{F}_3)$ lies in the kernel. For example, $H^2(X/G;\mathbb{F}_3)$ has the basis $\mu_{1,1}\sigma_{213}$, $\mu_{1,1}\sigma_{132}$, $\mu_{1,2}\sigma_{231}$, $\mu_{1,2}\sigma_{312}$, $\mu_{1,3}\sigma_{321}$ by the proof of Corollary 7.2.

9 The cyclic group of order q^2

It is natural to ask what happens for the regular representation of a larger cyclic q-group. We have not carried out this computation, but we can say where the argument of this paper stops. Let $G' = C_{q^2} = \langle g' \rangle$, let $K = \langle (g')^q \rangle$ be its subgroup of order q, let ψ be the character with $\psi(g') = \exp(2\pi i/q^2)$, and let $X' = \mathrm{Fl}(\mathbb{C}[G'])$. The fixed flags are again the coordinate flags, now indexed by S_{q^2}.

The first change is visible in Borel cohomology. The integral Chern class $c_1(\psi)$ generates $H^2(BG';\mathbb{Z}) \cong \mathbb{Z}/q^2$, and its reduction t' generates $H^2(BG';\mathbb{F}_q)$. Since $c_1(\psi^k) = kc_1(\psi)$, the class $c_1(\psi^k)$ reduces to kt', which depends only on k modulo q and vanishes when q divides k. The Chern roots of the dual of $\mathbb{C}[G']$ therefore run through each element of $\mathbb{F}_q t'$ exactly q times, and the relations in the Borel cohomology of X' come from

$$\prod_{k=0}^{q^2-1} (T + kt') = (T^q - (t')^{q-1}T)^q$$

instead of (5.2). The restriction of a class to a fixed flag depends only on the residues modulo q of the exponents of the eigenlines, in their order. Along a root curve whose weight ψ^k has $q \mid k$ the two restrictions are therefore equal, which is stronger than a congruence modulo t'.

These equalities have a geometric explanation in terms of the fixed locus of K. For $0 \le c < q$ let $V_c = \bigoplus_{j=0}^{q-1} \psi^{c+qj}$ be the eigenspace of K with character $\psi^c|_K$. A complete flag is fixed by K if and only if each of its subspaces is the sum of its intersections with the V_c. Such a flag determines a complete flag in each V_c and the order in which the dimensions of these intersections increase. So $(X')^K$ has $(q^2)!/(q!)^q$ components, each isomorphic to a product of q copies of $\mathrm{Fl}(\mathbb{C}^q)$. On V_c the generator g' acts as the scalar $\psi^c(g')$ times the regular representation of $G'/K \cong C_q$, and scalars act trivially on flags. Each component is thus a product of copies of the C_q-space X studied in this paper, with G' acting through G'/K. The root curves with $q \mid k$ lie in $(X')^K$.

In the genuine theory the difficulty appears already for single cells. The representation W_w of a cell can now contain characters ψ^k with $q \mid k$, on which K acts trivially; the cell of the longest permutation contains all of them. Suppose that a cell of complex dimension d contains $n > 0$ such characters. After restricting the action to K and smashing with H, its sphere becomes $\Sigma^{2n+(d-n)\lambda_K} H$, where λ_K is the realification of $\psi|_K$ and we use the results of this paper for the group K. This is not equivalent to $\Sigma^{d\lambda_K} H$: an equivalence would make u^n a unit of the coefficient ring of K, but $u\mu_{1,1} = 0$. The cells of length one, on the other hand, have the single faithful weight ψ^{-1}, and over K their spheres become $\Sigma^{\lambda_K} H$. After smashing with H the cells of X' are therefore not all suspensions by multiples of one representation of G'. The attaching maps then live in coefficient groups of G' in more general degrees, for which we have no analogue of Lemma 2.2(a).

The Euler classes behave differently as well. Write S^{ψ^k} and a_{ψ^k} for the sphere and the Euler class of the realification of ψ^k. The map $z \mapsto z^q$ from S^ψ to S^{ψ^q} is equivariant and is the identity

on fixed points, so a_{ψ^q} is a multiple of a_ψ in the coefficient ring of G'. The converse fails, because a_{ψ^q} restricts to zero on K and a_ψ does not. The ideal generated by a_{ψ^q} is thus strictly smaller than the one generated by a_ψ, whereas for G the Euler classes of all nontrivial characters generate the same ideal by (2.2).

We do not know whether $H \wedge X'_+$ splits as a wedge of representation suspensions of H, nor what the analogue of Theorem 1.2 should be. The description of $(X')^K$ suggests studying restriction to the G'-fixed flags in two stages, first restricting to $(X')^K$ and then using the action of G'/K on it, where the results of this paper apply to each factor.

A The transfer on the coefficient ring

We prove Lemma 2.4. It suffices to consider degrees in $\mathbb{Z} \oplus \mathbb{Z}\lambda$, since the units ε_r and their images $\bar{\varepsilon}_r$ are compatible with res and tr.

In these degrees $\underline{R}^\star(G/e)$ is spanned by the powers $\bar{u}^n$, $n \in \mathbb{Z}$, in degree $n(\lambda-2)$. For $n \geq 0$ the group $R^{n(\lambda-2)}$ is spanned by u^n, and for $n = -j < 0$ it is spanned by $\mu_{j,1}$. The degrees of $\nu_{j,k}$ and $\mu_{j,k}$ have underlying dimensions $1-2k$ and $2-2k$, so the $\mu_{j,1}$ are the only basis elements of $N_{1,\lambda}$ that can have a nonzero image under res. We therefore have to compute $\mathrm{res}(\mu_{j,1})$ and $\mathrm{tr}(\bar{u}^n)$.

Fix $j \geq 1$ and consider the cofiber sequence $S(j\lambda)_+ \to S^0 \to S^{j\lambda}$. Since the cohomology of a point vanishes in positive integer degrees at both orbits, the connecting homomorphism is an isomorphism of Mackey functors

$$\underline{H}_G^{2j-1}(S(j\lambda)) \xrightarrow{\cong} \widetilde{\underline{H}}_G^{2j}(S^{j\lambda}) = \underline{R}^{2j-j\lambda}.$$

The group G acts freely on the sphere $S(j\lambda) = S^{2j-1}$, with quotient the lens space $L = L^{2j-1}(q)$. For a free G-CW complex Y, the Bredon cochains with coefficients in $\underline{\mathbb{F}_q}$ are the cellular cochains of Y/G at the orbit G/G and those of Y at G/e. On cochains res is the pullback along $Y \to Y/G$, and tr sends a cochain on Y to the cochain on Y/G whose value on a cell is the sum of the values on its lifts. In integer degrees $\underline{H}_G^*(Y)$ is therefore $H^*(Y/G;\mathbb{F}_q)$ at G/G, as in [2, Proposition 3.2], and $H^*(Y;\mathbb{F}_q)$ at G/e, with res the pullback along the covering and tr its transfer in the sense of [13, §3.G]. For the covering $S^{2j-1} \to L$ both maps can be computed with integer coefficients in the top degree $2j-1$. The pullback is multiplication by the degree q of the covering. The composite of the transfer followed by the pullback is the sum over the deck transformations, which preserve the orientation, so it is also multiplication by q. Hence the transfer sends the orientation class of the sphere to that of the lens space. Reducing modulo q, the pullback $H^{2j-1}(L;\mathbb{F}_q) \to H^{2j-1}(S^{2j-1};\mathbb{F}_q)$ is zero and the transfer in the other direction is an isomorphism.

Consequently $\mathrm{res}(\mu_{j,1}) = 0$, which proves that res vanishes on N, and $\mathrm{tr}(\bar{u}^{-j}) = c_j\mu_{j,1}$ with $c_j \in \mathbb{F}_q^\times$. By Frobenius reciprocity and (2.6),

$$c_{j-1}\mu_{j-1,1} = \mathrm{tr}(\bar{u}^{1-j}) = u\ \mathrm{tr}(\bar{u}^{-j}) = c_j\mu_{j-1,1} \qquad (j \geq 2),$$

so all c_j are equal to a single scalar c. Rescaling every $\nu_{j,k}$ and $\mu_{j,k}$ by c preserves (2.6) and gives $\mathrm{tr}(\bar{u}^{-j}) = \mu_{j,1}$. For $n \geq 0$, reciprocity gives $\mathrm{tr}(\bar{u}^n) = u^n\,\mathrm{tr}(1) = qu^n = 0$. This is (2.9).

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